Robust Information Retrieval

SIGIR 2024 tutorial

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https://sigir2024-robust-information-retrieval.github.io/

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01:30–05:00

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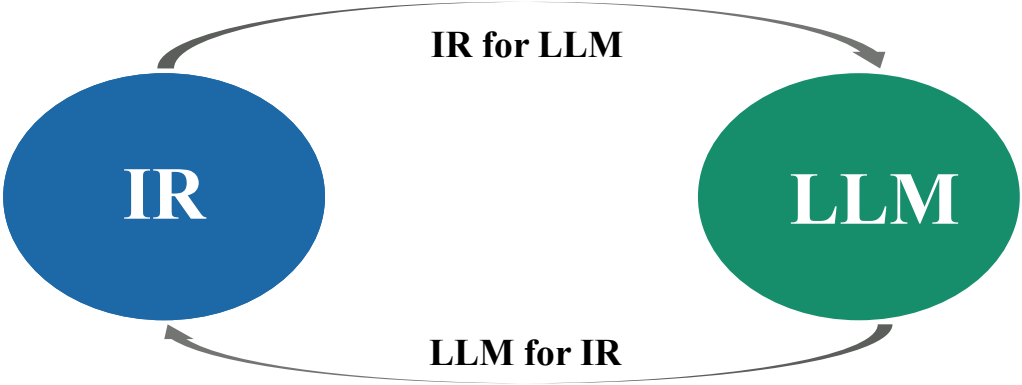
Section 5:
Robust IR in the age of LLMs

IR in the age of LLMs



- **IR for LLM**: Retrieval-augmented generation
- **LLM for IR**: A double-edged sword

New opportunities for IR robustness via LLMs

New opportunities to adversarial robustness

LLMs hold promise for improving the adversarial robustness of IR systems through their ability to generate and identify adversarial examples:

- **Generating adversarial examples with LLMs**
  - AIGC scenario
  - Superior capabilities in language generation and interaction
  - Hardening the IR system with generated adversarial samples
- **Adversarial defense assisted with LLMs**
  - Identifying adversarial samples
  - Enhancing existing defense strategies

New opportunities to OOD robustness

The powerful generation and language understanding capability of LLMs can help to improve the OOD robustness of IR systems:

- **Synthesizing OOD training data with LLMs**
  - LLMs can generate diverse and complex datasets that mirror OOD scenarios
  - Synthetic data can help improve the generalizability and robustness of IR models against OOD inputs
- **LLMs for OOD detection**
  - With capabilities of language understanding, LLMs can detect OOD queries
  - Neural IR models may perform worse on these OOD queries that deviate from the training distribution

New challenges for IR robustness via LLMs

New challenges to adversarial robustness

When applied to IR systems, the adversarial vulnerability of the LLMs themselves is introduced at the same time, as demonstrated by:

- **The vulnerability caused by hallucinations of LLMs**
- **Defense costs associated with the scale and opacity of LLMs**

New challenges to adversarial robustness

The vulnerability caused by hallucinations of LLMs

- With hallucination, LLMs can generate plausible yet factually incorrect information
- Such reliance can undermine the trustworthiness and reliability of the IR system

Prompt: Please **rank** the following documents according to their relevance to the query {{query}} and output the document IDs.
[1]{Doc_1}, [2]{Doc_2}, …, [n]{Doc_n}

Sure! I can help you. The relevance ranking is:
[2] > [3] > 358 > [1] ##3 > 68235 > ….

New challenges to adversarial robustness

Defense costs associated with the scale and opacity of LLMs

- LLMs operate as black boxes with limited transparency into how decisions are made
- This opacity complicates efforts to diagnose and mitigate vulnerabilities

New challenges to OOD robustness

LLMs have shown biases and input sensitivities in existing work, and these will affect the OOD robustness of IR systems:

- **Bias in the corpus domain of LLMs**
  - The training process of LLMs leads to a bias towards the domain characteristics
  - This can degrade performance when the model encounters OOD queries or documents
- **Sensitivity of LLMs to query inputs**
  - LLMs can exhibit high sensitivity to slight variations in input
  - This potentially leads to significantly different IR outcomes

So much to do . . .

Making robustness one of the hallmarks of IR in the age of LLMs!